Charan K

M.Tech in Artificial Intelligence

k.charankumar73@gmail.com | 9791239230 | linkedin.com/in/k-charan- | github.com/Charancode1

PROFESSIONAL SUMMARY

M.Tech Artificial Intelligence graduate with hands-on experience building LLM agents, RAG pipelines, and end-to-end ML systems. Proficient in Full stack web development and agentic AI frameworks. Experienced in designing and evaluating LLM-based systems, implementing ideas from recent NLP/ML research (NeurIPS, ICML, ACL), and deploying production-grade AI pipelines using Docker, Kubernetes, and AWS. Actively exploring agentic data engineering, LLM fine-tuning, and structured data reasoning with SQL and graph-based models.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, React, React Native, Node.js, Next.js, Nest.js

AI & ML: Large Language Models (LLMs), Agentic AI, RAG Pipelines, LLM Evaluation (Evals), Prompt Engineering, Computer Vision, Graph Neural Networks

Frameworks & Libraries: PyTorch, Hugging Face , TensorFlow, Keras, Flask, Fastify

MLOps & Tools: AWS, EKS, MLflow, Docker, Kubernetes, Streamlit, DynamoDB, Vector Databases

Other: REST APIs, Experiment Tracking, Model Deployment, Data Pipelines, Structured Data Reasoning

WORK EXPERIENCE

Software Engineer *Aug 2026 – Present*

PEPS-TECH Research and Development

- Design, develop, and deploy machine learning models for classification, detection, prediction, and automation tasks, and build ML pipelines in Python (NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch).
- Deploy and optimize ML models on Raspberry Pi, microcontrollers, and other embedded/edge platforms, building real-time applications that integrate sensors, cameras, and IoT modules.
- Owned end-to-end development of PEPS Atlas, from requirements, system architecture, database design, and development to production deployment and maintenance using Next.js, TypeScript, Supabase, and Cloudflare Workers.
- Managed data and infrastructure, including database migrations, data integration with external Supabase services, production environment configuration, secrets, security controls, and CI/CD pipelines for automated cloud deployments.
- Handled the complete product lifecycle, including debugging, production issue resolution, performance/reliability improvements, responsive UI/UX development, feature enhancements, and maintaining the application as a live production system.

AI Software Developer Intern *Nov 2025 – May 2026*

Home Ideas Technologies

- Designed and deployed Retrieval-Augmented Generation (RAG) pipelines using vector databases and LLMs for real-time AI-powered features, reducing query latency by integrating semantic search layers.
- Used Pinecone as the vector database for the Vigoe.ai conversational AI product, enabling efficient similarity search and retrieval for RAG-based features.
- Built agentic backend workflows using Node.js and Fastify that orchestrated LLM API calls, tool use, and structured data retrieval across web and mobile platforms.
- Deployed containerized AI applications on AWS using Docker and Amazon EKS, implementing scalable cloud-native infrastructure for production LLM services.
- Developed responsive web (React) and cross-platform mobile (React Native) interfaces integrated with AI backends, enabling end-to-end product delivery.
- Collaborated directly with the founding team on architecture decisions for AI system integration, covering data ingestion, model serving, and API design.

AI/ML Intern *July 2025 – Dec 2025*
LABMENTIX

- Developed and optimized machine learning and deep learning models for real-world use cases, improving model performance and reliability.
- Implemented computer vision and data analysis workflows using TensorFlow, PyTorch, and Keras.
- Built end-to-end ML pipelines covering data preprocessing, feature engineering, model training, experiment tracking (MLflow), and deployment readiness.

PROJECTS

Autonomous Research Agent with Self-Growing Knowledge Base

Python, LLMs, RAG, Hugging Face, Vector Databases, FAISS, Agentic Frameworks, APIs

- Designed a fully autonomous LLM agent that decomposes research objectives into sub-tasks using chain-of-thought reasoning and tool-use.
- Integrated live web and research APIs to collect, process, and structure unstructured data into a continuously expanding vectorized knowledge base with confidence-driven re-research loops. Implemented agent evals to monitor accuracy and reliability.
- Built modular, cloud-ready components with clean separation of agent logic, retrieval, and generation layers for scalable, real-world deployment.

GlaucoVision: Automated Glaucoma Detection Using GAN, U-Net, and VAE-CNN

PyTorch, Deep Learning, GAN, U-Net, VAE-CNN, Medical Image Processing

- Developed a hybrid deep learning system combining GAN-based data augmentation and U-Net segmentation for optic disc and cup detection from retinal images.
- Implemented a VAE-CNN architecture for feature extraction and glaucoma classification, reproducing and extending techniques from recent computer vision literature.
- Computed Cup-to-Disc Ratio (CDR) to enable accurate and early-stage glaucoma screening under limited labeled data conditions.
- Ran structured ablation experiments to evaluate each model component's contribution, documenting findings in a write-up format suitable for publication.

Voyage Analytics Platform – Cloud-Native ML Pipeline

Python, ML, MLflow, Flask, Docker, Kubernetes, Streamlit, SQL

- Built an end-to-end machine learning platform covering structured data analysis, SQL-based feature engineering, model training, experiment tracking, and inference.
- Managed model versions, metrics, and experiments using MLflow with a reproducible experiment design.
- Deployed models via Flask REST APIs, containerized with Docker, and orchestrated using Kubernetes (Minikube).
- Delivered a Streamlit-based analytics dashboard for interactive model performance visualization.

EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore *2024 – 2026*
M.Tech – Artificial Intelligence, CGPA: 7.31

Sathyabama Institute of Science and Technology *2020 – 2024*
B.E – Computer Science and Engineering, CGPA: 8.74

Sri Vijay Vidyalaya Matric Higher Secondary School *2020*
HSC, Percentage: 77%

Sri Vijay Vidyalaya Matric Higher Secondary School *2018*
SSLC, Percentage: 97.8%

RESEARCH PROJECT

Causally-Aware Temporal Graph Neural Network for Stock Influence Prediction

PyTorch, Temporal GNNs, Graph Attention Networks (GAT), GRU, Causal Inference, Granger Causality, Statistical Testing

- Developed a hybrid TGNN framework (GAT-GRU) to model dynamic stock interaction networks over 704 BSE stocks and 1,500+ trading days, designing the research question end-to-end from scoping to evaluation.
- Engineered a Temporal Gravity Model and Hybrid Influence Score to capture both structural and temporal dependencies in financial graph data.
- Applied Granger causality and F-tests to statistically validate causal influence beyond correlation, grounding model outputs in rigorous statistical testing.
- Presented research paper at the International Conference on Computational Intelligence and Sustainable Systems (CIS 2026), NIT Warangal (scrs.in/conference/cis2026); **to be published in the SCOPUS-indexed Springer Book Series, Lecture Notes in Networks and Systems (LNNS).**

CERTIFICATIONS

- Social and Economic Networks: Models and Analysis – Stanford University Online
- Responsible & Safe AI Systems – NPTEL
- Introduction to Large Language Models – NPTEL
- Geodata Processing Using Python and Machine Learning – ISRO
- 100 Days of Python Bootcamp – Udemy
- Data Science – ExcelR
- Full Stack Web Development – Udemy

HACKATHONS

RazorDefense – Autonomous Revenue Defense Platform

Razorpay AI Buildathon | Open Source | github.com/Charancode1/agentic-commerce-revenue-recovery

- Took part in the **Razorpay AI Buildathon** and built RazorDefense, an agentic commerce and revenue-recovery platform, which **was later open-sourced for the community**.
- Engineered autonomous agents that detect failed payments in real time and formulate bounded recovery strategies, deploying Gemini-powered shopper agents to win back at-risk revenue through Razorpay's Test Mode environment.
- Layered in policy guardrails, customer-consent workflows, real-time merchant monitoring, and tamper-evident audit trails, ensuring the autonomous system operated safely and transparently.